Physics-Guided Calibrated Localization of Methane Sources

Edward Cha, Eric Liang, and Alex Zhang

***Abstract*—We present a physics-guided method for localizing methane leaks from inexpensive sensor networks, with a statistical guarantee that the reported region contains the source 90% of the time. The core of the method is a set of *physics input images*: for every cell of a site map, closed-form formulas score how strongly the sensor readings point to a leak at that cell, how visible such a leak would be to the sensor layout, and how well the best-fitting leak rate at that cell explains the readings. Computed in milliseconds, these images are supplied to a standard neural network as additional input channels through a small zero-initialized head (0.2% added parameters), trained with ordinary labels, and calibrated with split conformal prediction. When the wind is only measured rather than known, the images are averaged over the wind sensor's stated error range, so the network also sees how fragile each cell's evidence is to wind error. On a testbed whose physics is exactly solvable, so the smallest 90%-mass region under the exact simulator posterior is computable, the method halves the median 90% region of a calibrated deep localizer from 94–98 m to 46–48 m across three network families, at the 50 m scale that U.S. EPA super-emitter notifications use to associate a leak with a facility. At a representative 10°/10% anemometer-error setting, where the evaluated measured-wind inversion baselines produce operationally uninformative calibrated regions, the method retains its guarantee at 70–73 m, with a quarter of cases below 50 m. All results are verified against exact references and reported over multiple training seeds.**

***Index Terms*—methane monitoring, conformal prediction, sensor networks, wind uncertainty, physics-guided machine learning**

I. Introduction

Methane traps roughly $80\times$ more heat than CO_2 over its first 20 years and fades from the atmosphere within about a decade, so stopping large leaks quickly is one of the highest-leverage climate actions available today. A large share of oil-and-gas methane comes from a small number of large point sources, known as super-emitters and defined by the U.S. EPA as events of $100\,\mathrm{kg\,h^{-1}}$ or more [2]. Detection technology has advanced rapidly; UNEP's satellite alert system issued over a thousand notifications in its first two years [1]. The bottleneck is what happens next: connecting an alert to a specific place that a crew can inspect. EPA's Super-Emitter Program (implementation extended to January 2027) asks notifications to identify any facility within **50 m** of the estimated leak location [2], and the EU imposes analogous source-level duties [3]. We therefore adopt 50 m as an operationally motivated target: a region small enough to trigger a focused inspection.

Continuous monitoring networks, in which a small number of inexpensive sensor masts capture the plume as the wind sweeps it across the site, are where this localization can happen. Machine learning provides instant estimates, and a standard statistical wrapper called split conformal prediction [4] adds a guarantee: a region that provably contains the true source 90% of the time. The gap we address is twofold. First, the guarantee speaks only to how *often* the region contains the source, not to how *small* the region is, and there was previously no practical way to measure how much smaller a guaranteed region could have been. Second, the classical alternative of inverting the plume physics directly requires the wind to be known exactly, while every real site has only a wind *measurement*; how each approach behaves under that everyday mismatch had not been carefully quantified.

Contributions. The paper centers on a single method and the evidence for it.

- **C1: Physics input images that carry their wind-error budget.** For every candidate cell, closed-form formulas produce four images: the evidence for a leak at that cell, the fragility of that evidence across the wind sensor's stated error range, the visibility of such a leak to the sensor layout, and the fit quality of the best leak rate at that cell. A zero-initialized head (0.2% added parameters) joins the images to any base network at the logits; training is ordinary supervised learning; split conformal prediction supplies the guarantee. When the error budget is set to zero, the images provably reduce to the classical matched-filter pair, so one recipe covers exact and measured wind alike.
- **C2: A measuring stick that quantifies the method's value.** A methane testbed (CH4-T) whose physics is exactly solvable, so the exact-model reference region is computable for every scenario and any guaranteed localizer can be audited in meters. A verified-90% deep localizer delivered 94–98 m where the reference is 6–7 m; the images close half of that gap on all three network families we tested (46–48 m).
- **C3: Robustness under measured wind.** At the evaluated 10°/10% anemometer-error setting, the measured-wind inversion baselines produce operationally uninformative calibrated regions (essentially the whole site), while the method degrades gracefully, and each added image tightens the guaranteed regions further, to 70–73 m.

Every component is standard and auditable, every number below comes from one shared, verified simulator with identical splits and calibration harness, and headline comparisons carry paired-bootstrap intervals over the same 2,000 test scenarios.

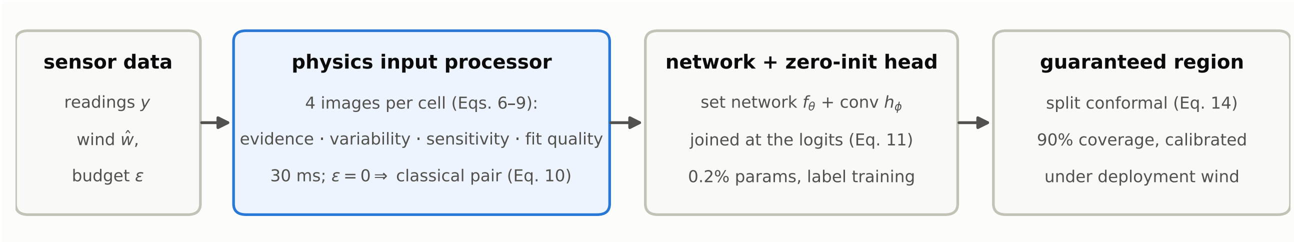

Fig. 1. The pipeline. The physics input processor turns raw sensor data and the wind sensor's stated error budget into four per-cell images (Eqs. 6–9), which any set network reads through a small zero-initialized head joined at the logits (Eq. 11); split conformal calibration certifies the 90% region under the deployment wind condition (Eq. 14). At ε=0 the images provably reduce to the classical matched-filter pair (Eq. 10).

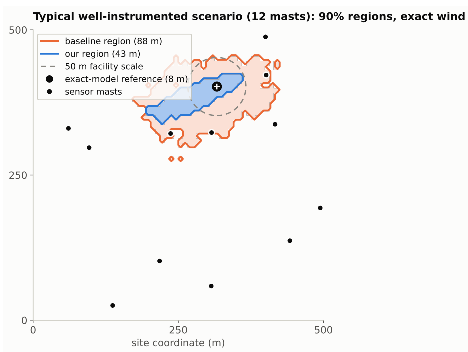

Fig. 2. Typical well-instrumented scenario: 90% regions of the labels-only baseline and of our method (conformal masks), the exact-model reference (equal-area circle), the sensor masts, and the 50 m facility scale. Selection rule: within the 10–12-mast subgroup, the scenario jointly closest (log scale) to that subgroup's median baseline and median method radii, i.e., typical for well-instrumented sites, not a best case.

II. Related Work

Conformal prediction gives distribution-free coverage guarantees [4]; how small the guaranteed sets are is the open half of the problem, and our audit scores it directly against the exact probability map at matched, verified coverage. **Simulation-based inference** trains neural networks to approximate probability maps from simulated data [6], [7]; its standard diagnostics check calibration [5], and recent work examines what happens when the simulator's settings differ from reality at deployment [8]. Our measured-wind design addresses exactly that setting by building the wind sensor's declared error range into the network's inputs. Our maps are hand-derived statistics, complementing learned summaries. **Source-term estimation** inverts dispersion physics with optimization or sampling at minutes to hours per event [9]; learned localizers answer in milliseconds, and our audit adds the missing instrument for judging how much information they extract. **Controlled-release testing** (METEC) evaluates continuous monitoring systems in the field [10]; our testbed mirrors that task structure with exact ground truth, complementing rather than replacing field validation.

III. Methodology

Our methodology has three layers, and each exists for a reason. A simulation system makes the truth knowable, so that every claim in this paper can be measured rather than assumed. Machine-learning localizers make inference fast and generalizable, answering in milliseconds on whatever sensor layout a site has. Between the two we insert a physics input processor, which computes the physical relationships the network would otherwise have to rediscover from data and presents them as inputs. Physics contributes exact understanding; learning contributes calibration and generalization; the sections below construct each layer in turn and state why it is built the way it is.

A. The simulation system: making the truth knowable (C2)

CH4-T places one source of unknown grid cell $c^* \in \{1, \dots, 64^2\}$ and rate q (10–500 kg h^{-1}, log-uniform, spanning the 100 kg h^{-1} super-emitter threshold) on a 500×500 m site watched by $S \in \{4, \dots, 12\}$ masts over T=30 one-minute steps. Transport follows the standard Gaussian plume with published Pasquill–Gifford/Briggs dispersion (verified against the original tables; mass conservation checked to 4×10^{-6}). In plume-frame coordinates (downwind distance x, crosswind offset y_{cw}, sensor height z_r, release height H), the concentration per unit rate is

$$g = \frac{1}{2\pi U \sigma_y(x)\, \sigma_z(x)}\, e^{-\frac{y_{\mathrm{cw}}^2}{2\sigma_y^2}} \left(e^{-\frac{(z_r-H)^2}{2\sigma_z^2}} + e^{-\frac{(z_r+H)^2}{2\sigma_z^2}} \right), \tag{1}$$

where U is wind speed and σ_y, σ_z are the stability-dependent dispersion widths. Stacking all sensors and time steps, a leak at cell c produces the fingerprint $g_c(w) \in \mathbb{R}^{ST}$ under wind sequence w, and readings are

$$y = q\, g_{c^*}(w) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{2}$$

Wind uncertainty is built into the data. Readings are generated under the true wind w, but the dataset records only the measurement $\hat{w}$, corrupted independently at every step t:

$$\hat{\theta}_t = \theta_t + \delta_t, \quad \hat{s}_t = s_t\, e^{\eta_t}, \quad \delta_t \sim \mathcal{N}(0, \sigma_\theta^2),\ \eta_t \sim \mathcal{N}(0, \sigma_s^2), \tag{3}$$

with $\sigma_\theta = 10°$ and $\sigma_s = 0.10$, the accuracy class of inexpensive monitoring anemometers. No model, at any stage, observes the true wind.

The simulator is deliberately built so that the truth is computable. Because the model (2) is linear in q, the per-cell likelihood depends on the data only through three summary quantities,

$$a_c = g_c^\top y, \qquad b_c = \|g_c\|^2, \qquad \rho = \|y\|^2, \tag{4}$$

and the exact probability map under known wind follows by integrating the rate out of the Gaussian likelihood:

$$p(c \mid y) \;\propto\; \int_0^\infty \exp\Big(-\,\tfrac{\rho - 2q a_c + q^2 b_c}{2\sigma^2}\Big)\,\pi(q)\,dq, \tag{5}$$

evaluated by high-precision quadrature (normalization checked to $\sim 10^{-12}$). **The exact-model reference region.** For each scenario we normalize (5) over cells, using a uniform source-cell prior and the same log-uniform rate prior $\pi(q)$ on $[10, 500]\,\mathrm{kg\,h^{-1}}$ used to generate the data (peak-aware composite Gauss–Legendre quadrature over that range), and apply the identical conformal wrapper of §III-D to the exact map itself, with the same randomized tie handling and strict threshold rule: the region is the smallest set of highest-posterior cells at the calibrated mass threshold. We call this the *exact-model reference region*. It is an oracle reference under the CH4-T simulator assumptions, not a universal information-theoretic lower bound. The two reported values correspond to different coverage levels: 7.0 m at nominal 90% (empirical 0.952; the exact map is conservative on the discrete grid) and 6.2 m at the learned models' matched empirical coverage of 0.89. A blocking check requires the wrapper applied to the exact map to reproduce its own guarantee before any experiment proceeds; this check caught two real implementation bugs, including a tie-handling correction that we report for reuse.

B. Base learners: why machine learning, and which networks

A monitoring system must answer in seconds, on whatever sensor layout a site has, under the wind it actually measured, while physics inversion takes minutes to hours per event and requires the wind to be known. A network trained once on simulated scenarios answers in milliseconds and, because it learns from examples, absorbs regularities with no closed form, such as how anemometer error distorts the evidence. The readings form a *set* of tokens (mast position, time, concentration, wind step) whose size varies from site to site, so the natural base learners are permutation-invariant set networks. We evaluate three families with different inductive biases, DeepSets (pooling), a graph neural network (relational structure between masts), and a Set Transformer (attention), plus a flow-based generative estimator, precisely so that the method's value is demonstrated independently of any one architecture choice.

C. The physics input processor (C1)

The plume model (1)–(2) fixes an exact relationship between any hypothesized leak and the readings it should produce, so testing a hypothesis is a correlation computation. The controlled comparisons of §IV-B show that networks trained end-to-end do not efficiently reconstruct this computation across 4,096 candidate cells (a correctly calibrated baseline delivers 94–98 m), while the computation by itself cannot judge how much to trust its own answer (calibrated alone, it returns the whole site). We therefore insert an *input processor* between the raw data and the network (Fig. 1): it evaluates the physics relationships explicitly, cell by cell, and presents the results as images aligned with the output grid, where a small convolutional head can read them spatially. This is the division of labor the paper centers on: physics supplies exact understanding of how a leak maps to readings, and the network supplies what physics lacks, calibration and generalization learned from examples. The classical matched-filter score of cell c normalizes the alignment a_c by the fingerprint's energy,

$$z_c \;=\; \frac{a_c}{\sigma\sqrt{b_c}}, \tag{6}$$

and $\log b_c$ records how visible a leak at c would be to the current sensor layout. These two $64{\times}64$ images are the zero-budget input maps. To carry the wind sensor's stated error range, we draw $K{=}8$ plausible true winds from the error model (3) centered on the measurement,

$$w^{(k)} \sim \mathcal{E}_\varepsilon(\hat{w}), \qquad k = 1, \dots, K, \tag{7}$$

recompute (4) under each draw, and feed four images: the average evidence and its variability,

$$\bar{z}_c = \frac{1}{K}\sum_k z_c^{(k)}, \qquad v_c = \Big(\tfrac{1}{K}\sum_k \big(z_c^{(k)} - \bar{z}_c\big)^2\Big)^{1/2}, \tag{8}$$

the average sensitivity $\overline{\log b_c}$, and the average *fit quality*. For each draw, the best nonnegative rate at cell c is $\hat{q}_c = \max(0, a_c/b_c)$, and the residual error it leaves is available in closed form,

$$R_c^{(k)} \;=\; \rho \;-\; 2\hat{q}_c a_c^{(k)} \;+\; \hat{q}_c^2 b_c^{(k)}, \qquad \bar{R}_c = \frac{1}{K}\sum_k R_c^{(k)}, \tag{9}$$

normalized per scenario (shifted to minimum zero, scaled by the median, clipped) so that all leak rates share one scale. Every image is closed-form linear algebra computed from the network's own inputs. Measured on an NVIDIA L40S at batch size one over 500 test scenarios after warm-up, host–device transfers included: preprocessing takes 29.6 ms median (31 ms p95) and the network forward pass 0.8 ms, for 30.4 ms median end to end. We verified (9) against direct reconstruction to relative error 10^{-7}.

Zero-budget reduction. With $\varepsilon = 0$ there is a single wind, so $v_c \equiv 0$, and the fit-quality image collapses onto the evidence image through the identity

$$R_c \;=\; \rho \;-\; \sigma^2 \max(z_c, 0)^2, \tag{10}$$

a fixed per-scenario transformation of z_c. The four images therefore carry exactly the information of the classical pair $(z_c, \log b_c)$: one recipe covers both settings, and the exact-wind results below are its zero-budget instantiation.

D. Training, the guarantee, and the division of labor

The input processor attaches to any base learner f_θ over the token set x through a three-layer convolutional head h_ϕ ($\sim$10k parameters, 0.2% of the model) that reads the map stack $M(x)$, joined at the output logits:

$$\ell(x) \;=\; f_\theta(x) + h_\phi\big(M(x)\big), \qquad \hat{p}(c \mid x) = \mathrm{softmax}\big(\ell(x)\big)_c, \tag{11}$$

with h_ϕ initialized at exactly zero, so at the first training step the model is the unmodified baseline and every comparison is controlled by construction. Training is ordinary supervised learning on smoothed one-hot targets $t(x)$:

$$\mathcal{L}(\theta, \phi) \;=\; -\sum_c t_c(x)\, \log \hat{p}(c \mid x). \tag{12}$$

Physics enters only through the inputs in (11).

For the guarantee we use split conformal prediction with a randomized highest-probability score. For a scenario with true cell c^*, the score is the probability mass ranked strictly above the true cell, plus a uniform share $u \sim \mathrm{Unif}[0,1]$ of the tied mass:

$$s(x, c^*) \;=\; \sum_{c:\, \hat{p}_c > \hat{p}_{c^*}} \hat{p}_c \;+\; u \sum_{c:\, \hat{p}_c = \hat{p}_{c^*}} \hat{p}_c. \tag{13}$$

The threshold $\hat{t}$ is the $\lceil (n{+}1)(1{-}\alpha) \rceil$-th smallest score over n calibration scenarios ($\alpha = 0.1$), and the region keeps the most probable cells, excluding only mass strictly below the threshold (the tie-handling correction of §III-A):

$$\mathcal{R}(x) = \big\{c : s(x, c) \le \hat{t}\,\big\}, \qquad \Pr\big[c^* \in \mathcal{R}(x)\big] \;\ge\; 1 - \alpha. \tag{14}$$

Calibration is always performed under the same wind condition in which the system is evaluated, so (14) certifies the deployment setting. Region sizes are reported as the radius of the circle of equal area; all models share identical data (10k/1k/2k/2k splits, a standard budget for this field [7]) and identical symmetry augmentation.

Scope of the guarantee. The conformal guarantee (14) is marginal over exchangeable scenarios drawn from the calibration distribution; it does not imply 90% coverage for every individual scenario, leak-rate range, sensor layout, or wind condition (subgroup diagnostics are summarized in Limitations). Calibration is performed separately for each evaluated deployment condition. The exact-model reference is used only to audit region efficiency under the synthetic testbed assumptions and is not part of the conformal guarantee.

Division of labor. The physics computes exact per-cell evidence through (6)–(9), knowledge a network would otherwise have to rediscover from data; the network, trained through (12), generalizes across layouts, rates, and wind histories and learns how much to trust each cell's evidence, knowledge with no closed form; conformal calibration (14) certifies the result.

Why not invert the physics directly? On this testbed one can; that is our exact-model reference (5). Deployed dispersion models have no exact solution, inversion takes minutes to hours per event [9], and §IV-C shows the evaluated inversion baselines are uninformative under measured wind.

TABLE I
THE AUDIT WITH EXACT WIND: MEDIAN 90% REGION RADIUS AGAINST THE EXACT-MODEL REFERENCE REGION; 2,000 TEST SCENARIOS.

Method	Coverage	Radius (m)
Exact-model reference region	0.89–0.95	**6.2–7.0**
Deep localizer, labels only	0.89	94–98
Ours (zero-budget setting)	0.89	**47.1–47.7**

TABLE II
EXACT WIND; 3 SEEDS PER CELL; COVERAGE 0.88–0.92 THROUGHOUT.

Estimator	Without maps	With maps
DeepSets	94–98	**47.1–47.7**
GNN	56–57	**46.2–47.5**
Set Transformer	69–71	**46.2–47.3**
Flow-based estimator (strengthened)	62.8–71.2	57.6–61.2
Classical score alone (no network)	—	282 (full site)

IV. RESULTS

A. The audit: a guarantee is not yet a useful region

Both learned rows pass every standard calibration check; only the audit reveals that one of them delivers regions more than ten times larger than the exact-model reference (Table I). Two input images and a small head close half of the distance. Relative to the labels-only model on the same 2,000 paired scenarios, the images reduce the median radius by 46.1 m (paired-bootstrap 95% CI $[43.6, 48.1]$), produce smaller regions in 96.4% of cases, and place 52.4% of regions below the 50 m facility scale (Clopper–Pearson 95% CI $[50.2, 54.6]$%); radius percentiles p50/p75/p90 are 48/86/131 m. The median case reaches the scale, and the upper tail marks the clearest room for future work.

B. The same fix works on every estimator we tested

Without the maps, the choice of network changes results by 40 m; with them, all nine runs of three network families land inside a 1.5 m band (Table II). The gain is available to whatever architecture a practitioner already has, with ordinary training and near-zero added cost. The flow-based estimator is informative in both directions: it improves on the baseline with no physics at all, and since we verified that its output layer can express 4.4 m regions, its remaining gap is not a matter of expressiveness; the images improve it as well (non-overlapping three-seed ranges), so the recipe composes with other estimators rather than competing with them. At the other extreme, the classical score alone identifies where the evidence lies but not how much to trust it, and pays for that overconfidence with full-site regions. Single-seed channel diagnostics point the same way: the evidence image alone reaches 65.0 m, the sensitivity image alone 98.5 m, both channels 47 m.

C. Measured wind: the main result

Real anemometers have error, and the effect on direct physics inversion is dramatic. Trusting the measured wind exactly, it concentrates its probability so far from the true

TABLE III

MEASURED WIND ($10°/10\%$ PER-STEP ERROR; NO METHOD SEES THE TRUE WIND; CALIBRATION UNDER THE DEPLOYMENT CONDITION; 3 SEEDS FOR LEARNED ROWS). MIDDLE ROWS USE PROGRESSIVELY LESS OF THE ERROR-RANGE INFORMATION.

Pipeline	Cov.	Radius (m)	<50 m
Direct physics inversion	1.00	282 (full site)	0%
tuned extra noise allowance	1.00	282 (full site)	0%
Network, no physics images	0.90–0.91	107.9–110.5	0%
random images, same head†	0.89	107.5	0%
Ours, single-wind images	0.91–0.92	82.2–84.2	15–17%
Ours, + avg. and variability	0.89–0.91	74.7–76.3	20–21%
Ours, + fit quality	0.90–0.92	**70.1–72.9**	**24–27%**

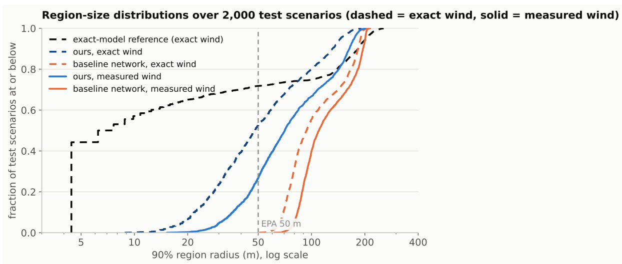

Fig. 3. Region-radius distributions over all 2,000 test scenarios (dashed: exact wind; solid: measured wind). The curves give medians, tails, and the fraction below the 50 m line in one view; our measured-wind distribution dominates the baseline's exact-wind distribution over most of its range.

source that, to keep its guarantee, the calibration wrapper must return essentially the entire site (Table III); allowing extra measurement noise, tuned on held-out data, does not change this. A negative control isolates the cause of our gains: feeding the identical head four *random* images of the same shape and scale (†, one seed, diagnostic) matches the no-physics baseline (107.5 m), so the improvement traces to the physics content of the images, not to added capacity or spatial regularization. The learned recipe behaves differently, and the pattern is clean: the more of the wind sensor's error-range information the input images carry, the smaller the guaranteed regions become, from 82–84 m using the measured wind alone, to 74–76 m after adding the moment images (8), to **70–73 m** after adding the fit-quality image (9). Relative to the no-physics model on the same 2,000 paired scenarios, the full method reduces the median radius by 39.4 m (paired-bootstrap 95% CI $[37.3, 41.9]$), produces smaller regions in 96.2% of cases, and places 26.6% of regions below 50 m (Clopper–Pearson 95% CI $[24.7, 28.6]\%$). Throughout, this is one network, one objective (12), one guarantee (14), one forward pass. The mechanism is constructive: because the physics and the sensor's own error range live in the inputs, ordinary supervised training gives the network thousands of examples from which to learn how the error behaves and how much to trust each cell's evidence. In summary: $94{\to}47$ m from supplying a missing computation; $47{\to}71$ m, the measured cost of imperfect wind; $108{\to}71$ m, the value of physics in the inputs at this setting.

V. LIMITATIONS

CH4-T is a measuring instrument, not a deployment: single steady sources, flat terrain, known stability class, and a $64{\times}64$ grid whose $\sim$7.8 m cells sit near the reference scale. The 6–7 m reference is a property of the testbed's assumptions, exists only under known wind (the measured-wind analogue is not computable, so 70–73 m is reported against the 50 m operational scale), and the wind error range is taken as known from the instrument specification; quantifying sensitivity to a misspecified range is a natural next step. With exact wind, roughly half of individual cases still exceed 50 m (p90 = 131 m); prespecified subgroup diagnostics show no collapsing group, with mild under-coverage for sparse-mast scenarios as the one consistent pattern (coverage 0.865–0.867 at 4–6 masts, n=690, about 3 standard errors below nominal), a concrete target for group-conditional calibration. Field validation on controlled releases is the natural next stage, and the comparisons reported here are exact regardless, because every method shares one verified simulator, one dataset, and one calibration harness.

VI. CONCLUSIONS

On a testbed with exact ground truth, a calibrated deep localizer delivered a tenth of the sharpness of the exact-model reference. Supplying one classical computation as input images brought three network families to 46–48 m at the 50 m facility-association scale, and the same recipe, carrying the wind sensor's stated error range in its inputs, retained its guarantee at 70–73 m at the evaluated anemometer-error setting, at 30 ms per scenario with 0.2% added parameters.

The research impact is a transferable evidence standard: exactly solvable testbeds as measuring instruments for learned inference, self-checking calibration gates, and one-change-at-a-time controls, applicable wherever guaranteed regions are produced. The social impact is practical: regions small enough to act on, produced in about 30 ms from inexpensive hardware, with stated guarantees and the wind sensor's imperfection budgeted honestly. This provides a controlled step toward converting methane detections into calibrated inspection regions.

REFERENCES

[1] UN Environment Programme (IMEO), "An eye on methane: Methane Alert and Response System annual report," 2024.

[2] U.S. EPA, "Standards of performance for GHG emissions: NSPS OOOOb/EG OOOOc and the Super-Emitter Program," 89 FR 16820, 2024; implementation timing per the 2025 Interim Final Rule.

[3] European Union, "Regulation (EU) 2024/1787 on methane emissions reductions in the energy sector," OJ L, 2024.

[4] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, 16(4), 2023.

[5] S. Talts *et al.*, "Validating Bayesian inference algorithms with simulation-based calibration," arXiv:1804.06788, 2018.

[6] G. Papamakarios and I. Murray, "Fast ε-free inference of simulation models with Bayesian conditional density estimation," *NeurIPS*, 2016.

[7] J.-M. Lueckmann *et al.*, "Benchmarking simulation-based inference," *AISTATS*, 2021.

[8] P. Cannon, D. Ward, and S. M. Schmon, "Investigating the impact of model misspecification in neural simulation-based inference," arXiv:2209.01845, 2022.

[9] T. Newman *et al.*, "Probabilistic inversion modeling of gas emissions," arXiv:2408.01298, 2024.

[10] C. Ilonze *et al.*, "Assessing continuous monitoring solutions under single-blind controlled testing," *Elementa*, 2025.